

Full Stack Developer Internship Assessment

Event Booking System

Overview

Build a full-stack Event Booking System that allows users to browse events and book seats.

The objective of this assessment is to evaluate your ability to design and develop a complete web application, including frontend development, backend APIs, application architecture, and overall code quality.

Tech Stack

You are expected to use:

Frontend

- React.js

Backend

- Node.js
- Express.js (recommended)

Database

- MongoDB / PostgreSQL / MySQL

You are free to use any additional libraries, frameworks, or tools that you find appropriate.

Requirements

Authentication

Users should be able to:

- Register
- Login
- Logout

Authentication should be implemented using a secure mechanism.

Event Management

List Events

Display all available events with:

- Event Name
- Description
- Date & Time
- Venue
- Total Seats
- Available Seats

Event Details

Users should be able to view detailed information about a specific event.

Booking Management

Book Seats

Users should be able to:

- Select an event
- Specify the number of seats to book
- Confirm booking

View Bookings

Users should be able to view all their bookings.

Cancel Booking

Users should be able to cancel an existing booking.

Cancelled bookings should release the booked seats back to the event inventory.

API Requirements

Your backend should expose APIs for:

Authentication

- Register User
- Login User

Events

- Get All Events
- Get Event Details

Bookings

- Create Booking
- Cancel Booking
- Get User Bookings

Design the APIs following REST principles and industry best practices.

Validation & Error Handling

Your application should:

- Prevent booking more seats than available
 - Validate user inputs
 - Handle invalid requests gracefully
 - Return meaningful error messages
-

Additional Enhancements

Feel free to improve the application beyond the requirements mentioned above.

We encourage candidates to showcase their engineering skills through thoughtful design decisions, application architecture, user experience improvements, testing, deployment, performance optimizations, or any other enhancements they consider valuable.

Submission

Submit a **single GitHub repository** containing:

- Frontend code
 - Backend code
 - Setup instructions
 - Project documentation
-

README Requirements

Your repository README should include:

Project Setup

Instructions to run the application locally.

Environment Variables

List all required environment variables.

API Documentation

Document the available APIs.

Assumptions

Mention any assumptions made during implementation.

Design Decisions

Briefly explain any significant architectural or technical decisions.

Submission Deadline

 **Monday, 11:00 PM IST**

Please submit your GitHub repository link before the deadline.

Only one submission per candidate will be considered.

Notes

- Focus on code quality, maintainability, and user experience.
- Write production-quality code wherever possible.
- Feel free to make reasonable product and engineering decisions.
- If any requirement is unclear, document your assumptions in the README.

Good luck, and we look forward to reviewing your submission.